Probability Problems & Solutions

A worked collection of probability interview problems

A set of 18 probability problems with full step-by-step solutions.

Problem 1 — Same-Side Probability

Statement

We have two coins. One is fair; the other is biased with P(heads) = 3/4. We select a coin at random and flip it twice. What is the probability that both flips show the same side?

Solution

Biased coin: P(same) = (3/4)(3/4) + (1/4)(1/4) = 9/16 + 1/16 = 10/16 = 0.625

Fair coin: P(same) = (1/2)(1/2) + (1/2)(1/2) = 1/2

Overall: P(same) = ½ × 0.625 + ½ × 0.5 = 0.3125 + 0.25

Final answer: 0.5625

Problem 2 — Expected Rounds to Form Teams

Statement

Six people split into two equal teams. Each round, every person shows a hand face-up or face-down (50/50, independent). If exactly three are face-up, they split into teams. What is the expected number of rounds before they split?

Solution

Each person is face-up with p = 0.5. Let F = number face-up; F ~ Binomial(6, 0.5).

$P(F = 3) = C(6,3)\,(1/2)^3\,(1/2)^3 = 20/64 = 0.3125$

Let R = number of rounds until success; R ~ Geometric(0.3125), so E[R] = 1/p.

Final answer: E[R] = 1 / 0.3125 = 3.2 rounds

Problem 3 — Ad Raters

Statement

People rate ads. 80% are careful and rate an ad good with probability 0.6 (bad with 0.4); 20% are lazy and rate every ad good. (a) With 100 raters each rating one ad, what is the expected number of good ads? (b) With 1 rater rating 100 ads, what is the expected number of good ads? (c) Given one ad was rated bad, what is the probability the rater was lazy?

Solution

Let G = "rated good", C = "careful rater".

P(G) = P(G|C)P(C) + P(G|C■)P(C■) = 0.6 × 0.8 + 1 × 0.2 = 0.68

(a) Number good ~ Binomial(100, 0.68); $E = np = 100 \times 0.68 = 68$

(b) $E = P(C)\cdot100\cdot0.6 + P(C$■$)\cdot100\cdot1 = 0.8\cdot60 + 0.2\cdot100 = 48 + 20 = 68$ (same as (a))

(c) A lazy rater never rates an ad bad, so P(bad | lazy) = 0. Hence P(lazy | bad) = 0.

Final answers: (a) 68 (b) 68 (c) 0

Problem 4 — First to Roll a Six

Statement

Amy and Brad take turns rolling a fair die; first to roll a 6 wins. Amy rolls first. What is the probability Amy wins?

Solution

P(Amy wins) = $1/6 + (5/6)^2(1/6) + (5/6)$■$(1/6) + \ldots$ (geometric series, ratio $(5/6)^2$)

$= (1/6) / (1 - (5/6)^2) = (1/6) / (11/36) = (1/6)(36/11)$

Final answer: 6/11 ≈ 0.545

Problem 5 — Jar of Coins

Statement

A jar holds 1000 coins: 999 fair and 1 double-headed. You pick a coin at random and toss it 10 times, getting 10 heads. (a) What is the probability the coin is double-headed? (b) What is the probability the next toss is also heads?

Solution

$P(F) = 0.999$, $P(D) = 0.001$, $P(10H \mid D) = 1$, $P(10H \mid F) = (1/2)^1$■

$P(10H) = 1 \times 0.001 + (1/2)^1$■ $\times 0.999 = 0.001 + 0.0009756 = 0.0019756$

(a) $P(D \mid 10H) = (1 \times 0.001) / 0.0019756 \approx 0.506$

(b) P(next heads) $= P(H|D)P(D|10H) + P(H|F)P(F|10H) = 1 \times 0.506 + 0.5 \times 0.494 \approx 0.753$

Final answers: (a) ≈ 0.506 (b) ≈ 0.753

Problem 6 — Three Increasing Cards

Statement

A deck of 500 cards numbered 1–500 is shuffled. You draw three cards one at a time. What is the probability each card is larger than the previous one?

Solution

Only the relative order of the 3 drawn cards matters; the population size is irrelevant.

There are 3! = 6 equally likely orderings, exactly one of which is strictly increasing.

Final answer: 1/6

Problem 7 — Second Card Not an Ace

Statement

You draw two cards one at a time from a shuffled 52-card deck. What is the probability the second card is not an ace?

Solution

$P = P(\text{1st ace, 2nd not}) + P(\text{1st not ace, 2nd not}) = (4/52)(48/51) + (48/52)(47/51)$

$= [48(4 + 47)] / (52 \times 51) = (48 \times 51)/(52 \times 51) = 48/52$

Final answer: $48/52 = 12/13 \approx 0.923$

Problem 8 — Four-Person Elevator

Statement

Four people ride an elevator in a building with 5 upper floors. Each picks a floor uniformly at random and independently. What is the probability that no two get off on the same floor?

Solution

$P(\text{all distinct}) = (\text{number of injective assignments}) / (\text{total assignments}) = (5\times4\times3\times2) / 5$■

$= 120 / 625$

Final answer: 0.192

Problem 9 — Is the Dice Game Worth Playing?

Statement

You roll two fair dice. If the sum is 7 you win \$21, but each roll costs \$10. Is the game worth playing?

Solution

$P(\text{sum} = 7) = 6/36 = 1/6$

Net profit: +\$11 (win, since 21 – 10) with prob 1/6; –\$10 (lose) with prob 5/6

$E[\text{profit}] = 11(1/6) - 10(5/6) = (11 - 50)/6 = -39/6 \approx -6.5$

Final answer: E[profit] ≈ –\$6.5 < 0, so the game is NOT worth playing.

Problem 10 — Five Heads out of Six

Statement

A biased coin shows heads 30% of the time. What is the probability of exactly 5 heads in 6 tosses?

Solution

$P(\text{5 heads}) = C(6,5)\ (0.3)$■ $(0.7)^1 = 6 \times 0.00243 \times 0.7$

Final answer: ≈ **0.0102**

Problem 11 — Three Zebras

Statement

Three zebras sit at the corners of an equilateral triangle. Each independently picks a direction (clockwise or counter-clockwise) along the triangle's edges and runs. What is the probability none collide?

Solution

They avoid collision only if all run the same way: all clockwise or all counter-clockwise.

P(all clockwise) = $(1/2)^3$ = 1/8; P(all counter-clockwise) = 1/8

Final answer: 1/8 + 1/8 = 1/4

Problem 12 — Raining in Seattle

Statement

You call 3 friends in Seattle and independently ask each if it's raining. Each tells the truth with probability 2/3 and lies with probability 1/3. All three say "Yes, it's raining." What is the probability that it is actually raining?

Solution

Let p = prior probability of rain. By Bayes' rule:

$P(\text{rain} \mid 3\times\text{Yes}) = P(3\times\text{Yes} \mid \text{rain})\cdot p \,/\, [P(3\times\text{Yes} \mid \text{rain})\cdot p + P(3\times\text{Yes} \mid \text{no rain})\cdot(1-p)]$

$P(3\times\text{Yes} \mid \text{rain}) = (2/3)^3 = 8/27$ (all three truthful); $P(3\times\text{Yes} \mid \text{no rain}) = (1/3)^3 = 1/27$ (all three lying)

$= (8/27)p \,/\, [(8/27)p + (1/27)(1-p)] = 8p \,/\, (7p + 1)$

Assuming an uninformed prior p = 1/2: $= 8(1/2) \,/\, (7(1/2) + 1) = 4 / 4.5$

Final answer: 8/9 ≈ 0.889

Correction note

A common shortcut gives 1 – $(1/3)^3$ = 26/27, but that actually computes the probability that at least one friend is telling the truth — not the probability that it is raining. The correct approach uses Bayes' rule and requires a prior probability of rain; with an uninformed prior of 1/2 the answer is 8/9.

Problem 13 — Netflix Lazy Raters

Statement

80% of raters are careful and rate 60% of movies good (40% bad); 20% are lazy and rate 100% of movies good. All raters rate equally many movies. What is the probability a movie is rated good?

Solution

P(good) = P(good | careful)P(careful) + P(good | lazy)P(lazy) = 0.6 × 0.8 + 1 × 0.2

Final answer: 0.68

Problem 14 — Secret Wins

Statement

100 students each toss a coin. Heads on the first toss = a real win. On tails, they toss again: heads on the second toss means they lie and claim a win; tails means they admit a loss. If 30 students claim they won, how many actually won?

Solution

P(claims win) = P(heads first) + P(tails then heads) = 0.5 + 0.25 = 0.75

P(really won | claims win) = P(really won) / P(claims win) = 0.5 / 0.75 = 2/3

Of the 30 who claimed a win, expected number who truly won = 30 × 2/3

Final answer: 20 students

Problem 15 — Marble Buckets

Statement

Bucket 1 has 30 red and 10 black marbles; Bucket 2 has 20 red and 20 black. A friend secretly picks one bucket and draws a marble that turns out red. (a) Probability it came from Bucket 1? (b) She returns it and draws again from the SAME bucket (with replacement); both draws are red. Probability both came from Bucket 1?

Solution

P(red | B1) = 30/40 = 3/4; P(red | B2) = 20/40 = 1/2; P(B1) = P(B2) = 1/2

P(red) = ½(3/4) + ½(1/2) = 3/8 + 1/4 = 5/8

(a) P(B1 | red) = (3/4 × 1/2) / (5/8) = (3/8)/(5/8) = 3/5

(b) Both draws are from the same bucket, so condition on "two reds" together:

P(RR | B1) = $(3/4)^2$ = 9/16; P(RR | B2) = $(1/2)^2$ = 1/4

P(RR) = ½(9/16) + ½(1/4) = 9/32 + 4/32 = 13/32

P(B1 | RR) = (9/16 × 1/2) / (13/32) = (9/32)/(13/32) = 9/13

Final answers: (a) 3/5 (b) 9/13 ≈ 0.692

Correction note

Because both marbles are drawn from the same (unknown) bucket, we must update on both reds jointly. Treating the draws as independent posteriors — $(3/5)^2$ = 9/25 — is incorrect; the correct probability is 9/13.

Problem 16 — Gambler's Ruin

Statement

You start with $30. Each fair coin flip wins $1 (heads) or loses $1 (tails). You play until you reach $0 or $100. What is the probability you reach $100?

Solution

This is the classic Gambler's Ruin with a fair, symmetric random walk, so the process is a martingale: E[final] = starting value = 30.

The walk ends at $100 (probability r) or $0 (probability 1–r): E[final] = 100r + 0(1–r) = 100r

Setting 100r = 30 gives r = 0.3

Final answer: 30%

Problem 17 — Fake Review Detection

Statement

98% of reviews are legitimate and 2% are fake. If a review is fake, the algorithm flags it as fake 95% of the time. If a review is legitimate, the algorithm correctly identifies it as legitimate 90% of the time. If the algorithm flags a review as fake, what is the probability it is actually fake?

Solution

P(fake) = 0.02, P(legit) = 0.98; P(flag fake | fake) = 0.95; P(flag fake | legit) = 1 – 0.90 = 0.10

P(fake | flagged) = (0.95 × 0.02) / (0.95 × 0.02 + 0.10 × 0.98) = 0.019 / (0.019 + 0.098) = 0.019 / 0.117

Final answer: ≈ 0.162 (about 16.2%)

Problem 18 — Biased Coin Among Many

Statement

A bag has 100 coins; 1 is double-headed (both sides heads), the rest are fair. You pick a coin at random and toss it 3 times, getting heads all three times. What is the probability the coin is the biased one?

Solution

Let A = "biased coin", B = "3 heads". P(A) = 1/100, P(A■) = 99/100, P(B|A) = 1, P(B|A■) = $(1/2)^3$ = 1/8

P(A | B) = (1 × 1/100) / (1 × 1/100 + 1/8 × 99/100) = (1/100) / [(1/100)(1 + 99/8)]

= 1 / (1 + 99/8) = 1 / (107/8) = 8/107

Final answer: 8/107 ≈ 0.075